

Michael Chen

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EDUCATION

University of Washington

Master of Science in AI/ML for Engineering

Expected 2028

University of Illinois Urbana-Champaign

Bachelor of Science in Physics – Computational Physics Professional Track

May 2026

- **GPA:** 3.75
- **Relevant Coursework:** Machine Learning in Physics, Quantum Computing, Data Structures, Numerical Methods, Advanced Laboratory

EXPERIENCE

IDX Exchange

Data Analyst Intern

Summer 2026

- Built **Python data pipelines** to clean, validate, transform, and analyze monthly real-estate MLS listing and transaction datasets for downstream market and competitive analytics.
- Performed exploratory data analysis and data-quality validation using **pandas, NumPy, and statistical methods**, identifying missing values, invalid records, inconsistent dates, and price-related outliers.
- Engineered analytical features including **price per square foot, close-to-list ratios, days on market, contract timelines, and monthly transaction metrics** for housing-market analysis.
- Integrated external mortgage-rate data and geospatial school-district information with MLS records using **data joins, geographic coordinates, and GeoPandas spatial joins**.
- Developed interactive **Tableau dashboards** for housing-market and competitive analysis, including price trends, sales volume, listing agents, offices, and ZIP-code-level geographic analysis.

MACHINE LEARNING & DATA SCIENCE PROJECTS

Galaxy Zoo Galaxy Classification | *Python, PyTorch, CNNs, NumPy*

- Developed a **convolutional neural network** in PyTorch to classify galaxy morphology from astronomical image data using the Galaxy Zoo dataset.
- Built an image-processing and model-training workflow involving tensor transformations, convolutional layers, pooling, nonlinear activations, optimization, and model evaluation.
- Analyzed model performance and classification behavior to understand how learned visual features distinguish different galaxy structures.

Hopfield Networks & Restricted Boltzmann Machines | *Python, NumPy, Machine Learning*

- Implemented **Hopfield neural networks and Restricted Boltzmann Machines from scratch** using Python and NumPy without high-level machine-learning frameworks.
- Implemented core learning and inference algorithms including network-state updates, energy-based optimization, probabilistic sampling, and weight updates.
- Applied the models to associative-memory and generative-learning tasks, developing a deeper understanding of the mathematical foundations of neural networks.

Quantum Computing Simulator & Shor's Algorithm | *Python, NumPy, Quantum Computing*

- Built a quantum-computing simulator in Python to model quantum states, gates, measurements, and multi-qubit operations using linear algebra.
- Implemented components of **Shor's algorithm** and analyzed the computational principles behind quantum period finding and integer factorization.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Generative Models, Energy-Based Models

Data Science: pandas, NumPy, Matplotlib, GeoPandas, Data Cleaning, Feature Engineering, Exploratory Data Analysis, Statistical Analysis

Data & Visualization: Tableau, Geospatial Analysis, Excel

Developer Tools: Git, GitHub, Jupyter, VS Code